# Ceferino Patino

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## Objective

Aspiring computer scientist eager to continue learning and develop practical skills through meaningful industry experience. Seeking opportunities to apply my software development knowledge while contributing to innovative projects.

## Education

**B.S. in Computer Science** — Lincoln, NE, United States
*University of Nebraska – Lincoln* — *May 2024 – May 2026*

## Honors and Awards

**Undergraduate Student Researcher Award** — University of Nebraska – Lincoln, *2025*

**Dean's List** — University of Nebraska – Lincoln, *Fall 2024 - Spring 2026*

## Skills

**Languages**: Python, C++, Rust, JavaScript, Lua, Bash, SQL, Nix, Go, Java

**Frameworks & Libraries**: PyTorch, OpenCV, Astro, React, Express.js

**Tools & Platforms**: Docker, K8, Terraform, Git, Nix/NixOS, Maturin, MPI, HDF5, AWS, Django, PostgreSQL, MySQL, sqlite

**Concepts**: Machine Learning, RL/MARL

**Software Engineering**: SOLID, Agile, SCRUM, CI/CD, DevOps, Parallel Programming

**Soft Skills**: Team collaboration, Technical communication, Self-directed learning, Mentorship

## Certifications

**AWS Certified Cloud Practitioner** — Amazon Web Services, *October 2023 – May 2026*

## Experience

**Mutual of Omaha** — Remote
*Application Developer ← I/S Intern* — *May 2026 – October 2026*

- Developed and maintained enterprise applications using Java and Groovy, building RESTful APIs and backend services to support business operations
- Built responsive front-end interfaces with Angular, JavaScript, and TypeScript, improving user experience for internal stakeholders
- Designed and optimized complex queries in SQL and Oracle SQL to retrieve and manipulate data across multiple database systems
- Collaborated with cross-functional teams to design and implement system enhancements that improved operational efficiency
- Maintained and integrated existing applications while ensuring compliance with company Information Services policies and security standards
- Implemented appropriate security and privacy measures to protect sensitive company and customer data

**OASYS Lab – University of Nebraska – Lincoln** — Lincoln, NE, United States
*Undergraduate Researcher* — *May 2024 – August 2026*

- Contributed to advanced MARL research in a graduate research group, developing a benchmark suite used by 20+ researchers internationally.
- Enabled dynamic agent/task/frame adaptation for open-system evaluation; used by 20+ international researchers.
- Developed `free-range-rust`, a Rust+CUDA backend to accelerate `free-range-zoo`, doubling environment performance.
- Integrated Python bindings with Maturin to expose custom CUDA logic for seamless use in RL workflows.
- Co-led MOASEI competition at AAMAS 2025 with 10+ teams, benchmarking MARL agents in open-agent/task settings

**Raman Lab – Washington University** — Remote
*Software Developer* — *June 2024 – May 2026*

- Developed real-time imaging software for a 3D 2-photon lightsheet microscope processing 20 GB/s of image data.
- Designed parallel MPI/HDF5 data acquisition and OpenCV pipelines for volumetric dataset processing.

**CSCE361 Software Engineering – University of Nebraska – Lincoln** — Lincoln, NE, United States
*Undergraduate TA under Dr. Bhuvana Gopal* — *August 2025 – May 2026*

- Support students in labs, assignments, and group projects for CSCE361, a course focused on large-scale software development and engineering best practices.
- Host technical walkthroughs and office hours to assist with programming and technology tools.
- Grade quizzes and exams, and provide feedback on course deliverables.
- Facilitate team-based learning and help students understand software engineering concepts, including requirements analysis, design, testing, and maintenance.

**Danforth Plant Science Center** — St. Louis, MO, United States
*Software Developer* — *December 2023 – August 2024*

- Developed computer vision software for analysis of underground agricultural data for plant phenotyping.
- Worked with developer teams to manage AWS infrastructure for processing high-density aerial crop data.

## Open Source Contributions

**nixpkgs** NixOS Project
*Contributor & Package Maintainer* *May 2025 – Present*

- Contribute to nixpkgs with improvements, reviews, and new package additions
- Collaborate with the community to ensure package quality and build reproducibility

**nixvim** NixOS Project
*Maintainer* *May 2025 – Present*

- Maintain Nixvim, a modular Neovim compatibility port for Nix
- Add plugin support for popular vim plugins and neovim integrations and improve docs for the community

**Turso** Distributed Edge Database
*Contributor & Package Maintainer* *2025*

- Fixed a core bug in Turso, a distributed edge database built on libSQL, improving reliability for production users
- Collaborated with maintainers to review, test, and merge the fix into the main codebase
- Engaged with the open-source community to ensure robust, high-quality contributions

## Projects

**cpatino.com** Personal
*Web Development – JavaScript, Astro* *May 2025 – Present*

- Built and deployed a personal site with Astro, optimized for minimal bundle size and fast loads

**yumevim – "dream vim"** Personal
*Dev Tooling – Lua, Nix, Neovim* *July 2024 – Present*

- Architected a modular, declarative Neovim configuration with Lua and Nix for reproducible setups across devices
- Implemented zero-downtime updates with atomic rollbacks using Nix flakes and Git

**3D Volumetric 2-Photon Lightsheet Microscope** Raman Lab – Washington University
*Biotechnology / Microscopy – C++, MPI, HDF5, OpenCV* *June 2024 – Present*

- Developed software for a cutting-edge 3D volumetric 2-photon lightsheet microscope—one of few worldwide
- Built parallel data acquisition pipelines with MPI and HDF5 handling 20 GB/s of image data
- Optimized imaging workflows with OpenCV for real-time processing of volumetric datasets

**free-range-zoo** OASYS Lab – University of Nebraska – Lincoln
*AI / Reinforcement Learning – Python, PyTorch* *April 2024 – Present*

- Developed open-environment benchmarks for POSG-based multi-agent RL (wildfire, cybersecurity, rideshare)
- Implemented vectorized training loops for MADDPG, COMA, and GNN-based RL policies
- Enabled dynamic agent/task/frame changes to benchmark adaptability in open systems
- Used by 20+ researchers internationally as a benchmark RL algorithms in multi-agent environments

**free-range-rust** OASYS Lab – University of Nebraska – Lincoln
*AI / Reinforcement Learning – Rust, CUDA, Maturin* *April 2024 – Present*

- Implemented custom CUDA kernels in Rust to accelerate dynamic RL environments
- Doubled performance of `free-range-zoo` through low-level vectorized space operations
- Exposed Rust/CUDA modules to Python via Maturin for seamless integration and use by other research groups

**yumeami – "dream network"** Personal
*DevOps – Nix, NixOS* *April 2024 – Present*

- Engineered modular NixOS configurations with separate system and home profiles across heterogeneous hardware
- Developed a self-hosted infrastructure integrating services, CI/CD pipelines, and custom tooling for automation
- Implemented declarative provisioning and deployment workflows enabling reproducible, scalable environments

**UNL Senior Design Capstone (Mutual of Omaha – Sponsor)** Lincoln, NE, United States
*Student Software Engineer* *September 2025 – May 2026*

- Co-led design and development of a fullstack JSON validation engine for Mutual of Omaha internal workflow automation.
- Built and deployed the solution using React (frontend), Spring Boot (API backend), AWS CDK for infra-as-code, and ECS/Aurora for production cloud delivery.
- Collaborated with Mutual of Omaha engineers to deliver requirements, review code, and build CI/CD pipelines for cloud deployment.

**Fonio Seed Computer Vision** Donald Danforth Plant Science Center
*Computer Vision – Roboflow, OpenCV* *April 2024 – August 2024*

- Compiled and labeled a dataset of over 8,000 fonio seed images for robust model training
- Implemented a neural network achieving 99% accuracy in detecting overlapping seeds and providing precise counts

**SubTerra.ai** Donald Danforth Plant Science Center
*Computer Vision / Agricultural AI – PyTorch, OpenCV, Django, Docker* *December 2023 – August 2024*

- Developed a Django interface to run segmentation and extrapolate root volume
- Built root segmentation models using PyTorch for high-resolution root image data
- Processed segmentation masks with OpenCV to compute volumetric estimates achieving 97% accuracy

**FieldDock** Donald Danforth Plant Science Center
*Computer Vision / Agricultural AI – OpenCV, Docker, AWS* *December 2023 – May 2024*

- Deployed multiple components of a high-throughput drone data collection system using Docker and AWS
- Developed flight software for drone vision and landing leveraging OpenCV

**qOverflow** Black Data Processing Associates (BDPA)
*Web Development – JavaScript, MongoDB, React, Express.js, Node.js* *June 2022 – August 2022*

- Won first place in 2022 BDPA hackathon developing a Stack Overflow-inspired Q&A platform with custom analytics
- Containerized the app using Docker and deployed on Kubernetes; documented APIs with SwaggerHub

**hypixel-helper** Personal
*Web Development – JavaScript, React, Redis* *April 2022 – May 2022*

- Built live tracking and forecasting of Hypixel Skyblock marketplace data using SARIMA

## Publications

**Inaugural MOASEI Competition at AAMAS'2025: A Technical Report** arXiv:2507.05469
*Patino C., Billings T., Abadi A., Redder D., Eck A., Doshi P., Soh L.* *2025*